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Performance Comparison of BLDC Motor Controllers Designed Based on Trapezoidal Commutation and FOC

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Abstract. A motor controller is an important component in an electric vehicle. Research and development activities of brushless direct current (BLDC) motor controllers are devoted to getting better performance and higher efficiency. This topic still attracts researchers' interest all over the world, as can be seen from the amount of new scientific publication. BLDC motors in electric vehicle applications require transient performance improvement during start-up. PUI SKO ITS has been conducting research and development on BLDC motor controllers with trapezoidal commutation and field-oriented control (FOC) types. This paper presents an experimental performance comparison between the two BLDC controllers. Experimental results show that the trapezoidal controller is more responsive than the FOC controller, but the FOC controller has a smoother speed response than the trapezoidal controller. Moreover, the motor controller based on FOC produces smaller torque overshoot and smaller steady-state torque than the motor controller based on the trapezoidal commutation. This result calls for deeper analysis to push the performance of both types of the BLDC controllers to their limits and for further comparison in a possible higher stage of each controller's performance.

INTRODUCTION

The rapid development of electric vehicles in the world has led to the increasing need for brushless direct current (BLDC) motor controllers with the latest technology in the industry. A BLDC controller is needed that has high efficiency at various speeds, good stability, and has good transient performance [1]. Modern control methods have several types such as optimal control, variable structure control, nonlinear control, and controls that are generally used to control the speed of the permanent magnet motor [2]. However, this approach is complex in the theoretical foundation [3]. Trapezium controller is the easiest and most efficient solution for many control problems in the world because of its functionality which includes transient and steady state responses [4]. The trapezoidal controller is more often applied because it has a simple structure, high reliability, and is easy to implement [5]. But the trapezoidal controller has the disadvantage of turbulent torque delivery.

BLDC motor controller with high performance generally uses vector control. Paper [6] reported a compare between field-oriented control (FOC) and traditional brushless DC motor controller with proportional-integral (PI). The FOC has less torque ripple than the traditional brushless DC motor controller with PI. Speed performance on the FOC controller increases compared to the traditional brushless DC motor controllers with PI. Paper [7] proposed the FOC PMBLDC motor using SV-PWM. The reference voltage ($V_d \text{ ref}$, $V_q \text{ ref}$) is generated from the current error signal ($I_d \text{ error}$, $I_q \text{ error}$) using an adaptive PI controller. Paper [8] proposes FOC PMBLDC motors using rectangular current

regulated PWM. But FOC has a major problem with the input parameters. If the inputted parameters are not suitable it will cause efficiency to get worse compared to the traditional controller, it can even cause failure to the controller.

Center of excellence in science and technology (PUI) of Automotive Control System (SKO) - ITS has been conducting research and development on BLDC motor controllers with trapezoidal commutation and FOC types. This paper compares the performance of both types of BLDC controllers. The two motor controllers will be installed with the same 30kW BLDC motor to determine the performance comparison of two controllers. From the testing, information was obtained in the form of power, torque, and acceleration time as a comparison point between two controllers. Thus it will be able to determine the right type of controller according to this application. This paper is organized as follows. Trapezoidal controller and FOC controller are overviewed in section II. Experimental comparison of the two controllers is explained in section III. Finally, the conclusion is drawn in section IV.

TRAPEZOIDAL CONTROLLER AND FOC CONTROLLER

Motor-driven electronically can also be called the BLDC motor. In the BLDC motor there is no brush on the rotor, the commutation is done electronically. In general, the stator made of magnetic steel sheet composition. The magnetization of the permanent magnet and its rotation on the rotor are adjusted so that the resulting back emf is trapezoidal. This causes a rectangular three-phase voltage system that is used to create rotational fields with less torque ripples.

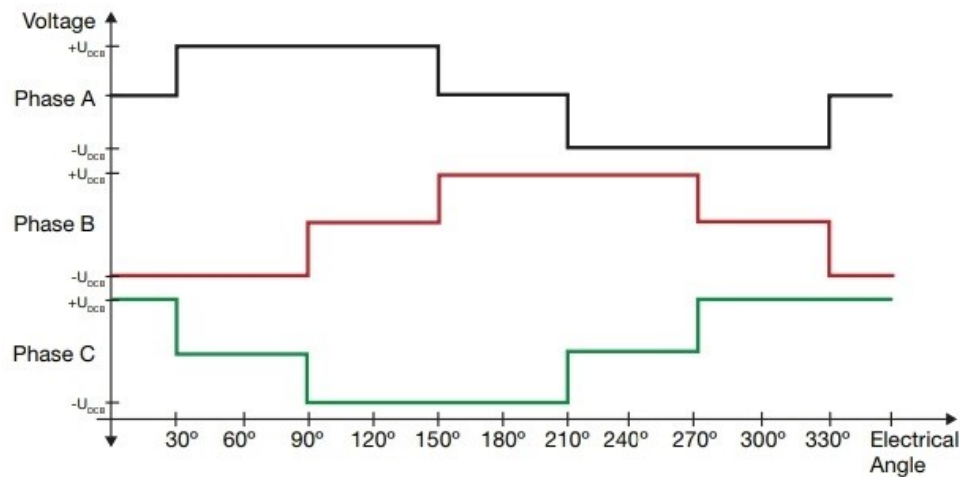


FIGURE 1. Six step commutation[10].

In general, BLDC motor rotation control uses computational six steps control techniques (60 degrees or 120 degrees). In the six step control produces a voltage with six vectors in each one electronic rotation shown in figure 1. Back EMF must be aligned with the amplitude and phase of the applied voltage. The BLDC controller must be able to control the amplitude used and align the six-step commutation with the rotor position. At a certain angle of the rotor position must be known, to synchronize between the voltage applied to the back EMF Under that condition, the BLDC motor is controlled with less torque ripple and power efficiency.

Figure 2(a) shows a 3 phase inverter that has 3 half-bridge switch devices. Each switch device has a label Q_1 , Q_2 , Q_3 , Q_4 , Q_5 , and Q_6 . Each switch is activated based on the position of the rotor following six-step commutation. Fig. 2(b) shows the equivalent model of a BLDC motor.

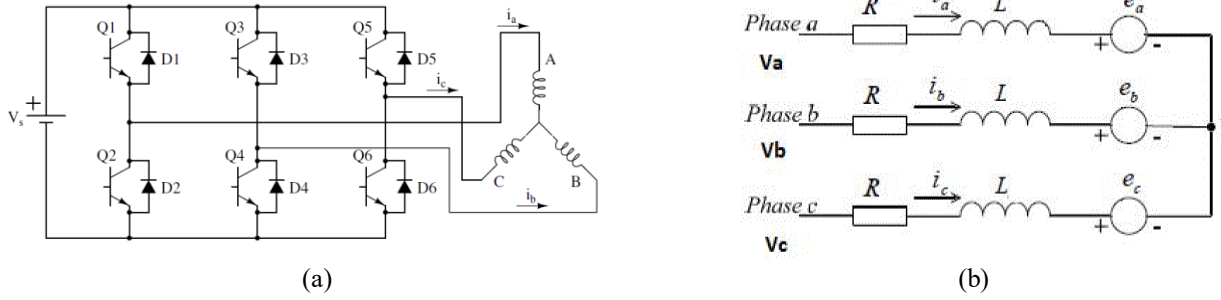


FIGURE 2. Three-phase star connected BLDC motor: (a) 3 phase inverter; (b) equivalent model BLDC motor [9].

The three-phase star-connected BLDC motor explained by following the equations[9]:

$$v_{ab} = R_s(i_a - i_b) + L_s \frac{d}{dt}(i_a - i_b) + e_a - e_b \dots \dots \dots (1)$$

$$v_{bc} = R_s(i_b - i_c) + L_s \frac{d}{dt}(i_b - i_c) + e_b - e_c \dots \dots \dots (2)$$

$$v_{ca} = R_s(i_c - i_a) + L_s \frac{d}{dt}(i_c - i_a) + e_c - e_a \dots \dots \dots (3)$$

$$T_e = T_L + B\omega_m + J \frac{d\omega_m}{dt} \dots \dots \dots (4)$$

Where: v = voltages, i = currents, and e = phase back-emf in the three phases a , b , and c . The resistance (R) and inductance (L) are the values in each phase, while T_e and T_L are the torque of electric and load. Inertia of the rotor (J), friction constant (B) and speed of the rotor (ω_m).

The above dynamical equations of the BLDC motor represent a time-variant system with voltage, current, back-emf, and magnetic flux are time-varying quantities. To ease the analysis of the electric machine, mathematical transformations are utilized to decouple variables and to facilitate the solution of a time-variant system. The stator phases use instantaneous currents (i_a , i_b , and i_c), the value of stator current $\vec{I}_s$ can be known using the following equation (5)[11].

$$\vec{I}_s = i_a + i_b e^{j2\pi/3} + i_c e^{j4\pi/3} \dots \dots \dots (5)$$

Where: a , b , c are the axes of the three-phase system.

The current vectors can be represented as sinusoidal three phases which need transform into an in-variant coordinate in two axes. The steps of transformation:

1. $(a, b, c) \rightarrow (\alpha, \beta)$ (the Clarke transformation), provides the output of the variants of the time coordinate.
2. $(\alpha, \beta) \rightarrow (d, q)$ (the Park transformation), which provides the output of the variants of the time coordinate.

The quantity of three-phase voltage/current (axis a , b , and c) transformed into two-phase of voltage/current. The a -axis in the same direction with and α axis, but orthogonal with β -axis, its shown in Fig. 3 which illustrates the Clarke transformation. The three-phase system will modifies into two orthogonal dimension (α, β) system as expressed in (6) and (7)[12].

$$i_{s\alpha} = i_a \dots \dots \dots (6)$$

$$i_{s\beta} = i_a/\sqrt{3} + 2i_b/\sqrt{3} \dots \dots \dots (7)$$

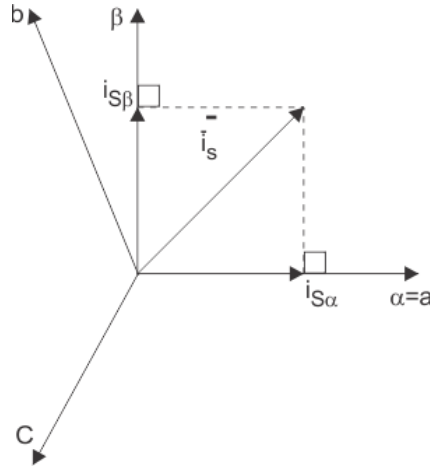


FIGURE 3. Vector diagram of Clarke transformation[12].

Figure 4 illustrate the Park transformation. Axis d and q are projections modified from axis α and β using rotating system. Notice that θ is the flux position of rotor.

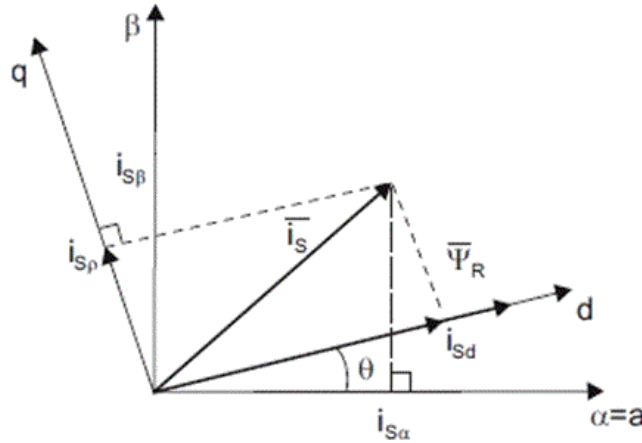


FIGURE 4. Vector diagram of Park transformation[12].

The torque and flux vector components are currently determined by the following equation[12]:

$$i_{sq} = i_{s\alpha} \sin \theta + i_{s\beta} \cos \theta \dots \dots \dots (8)$$

$$i_{sd} = i_{s\alpha} \cos \theta + i_{s\beta} \sin \theta \dots \dots \dots (9)$$

Current vector (α , β) and rotor flux position is using on this components. The position of the rotor flux accurately shown the calculated vector (d , q) components easily. This is also control the torque directly, because the flux (i_{sd}) and torque (i_{sq}) independent. This leads to a control method known as field-oriented control (FOC).

EXPERIMENTAL PERFORMANCE COMPARISON

Controller algorithm was tested using a chassis dynamometer. Performance tested will be analyzed such as characteristic power, torque, and time acceleration also we can generate load from this dynamometer. This paper tested a controller of BLDC motor using the trapezoidal control algorithm and the FOC algorithm.

Figure 5(a) shows the experimental set-up. A BLDC motor with 30 kW of power is connected to a dynamometer. For safety reason, during the experiment, the BLDC motor is limited to 6 kW of power. Figure 5(b) shows the human-

machine interface of the dynamometer. The human-machine interface displays a form of power, torque, and speed (rpm). Table 1 lists up the specification of the BLDC motor used in this experiment.

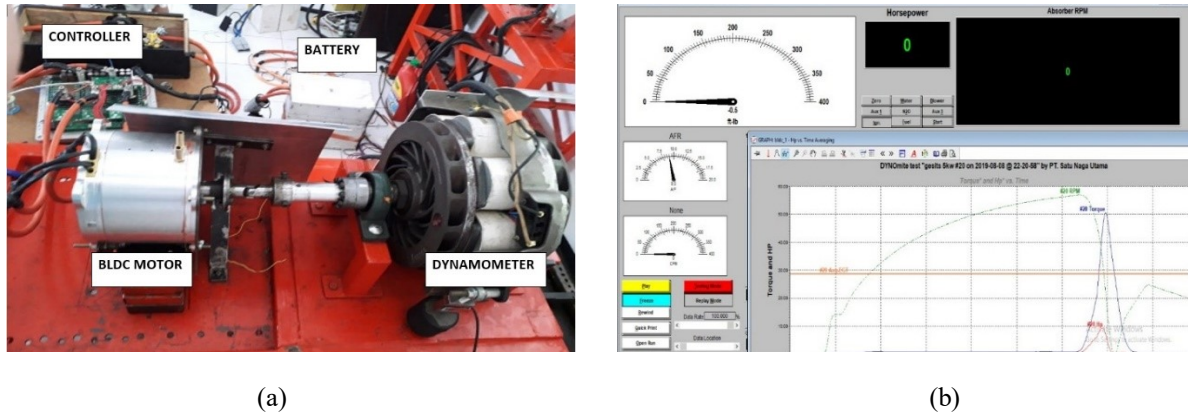


FIGURE 5. Experimental set-up: (a) The performance test scheme; (b) The human-machine interface of the dynamometer.

TABLE 1. BLDC MOTOR SPECIFICATION

Specification	Value	Unit
Motor Power	30000	Watt
Operation Voltage	100	VDC
Maximum Current	240	Ampere
Rated Speed	7000	Rpm
Resistance (R)	8.005	$m\Omega$
Inductance (L)	158.485	mH
Number of Poles	4	Pole Pair

Figures 6 and 7 show the experimental results. The motor rotor was coupled with the dynamometer rotor. Initially, the motor was rotated by the controller up to the maximum speed, then a load torque was applied through the dynamometer until the motor stopped rotating. Fig. 6 shows the history of the motor rotor speed time, the dynamometer load torque, and the power recorded by the dynamometer. The green lines are the results using the trapezoidal controller while the blue lines are the results using the FOC based controller.

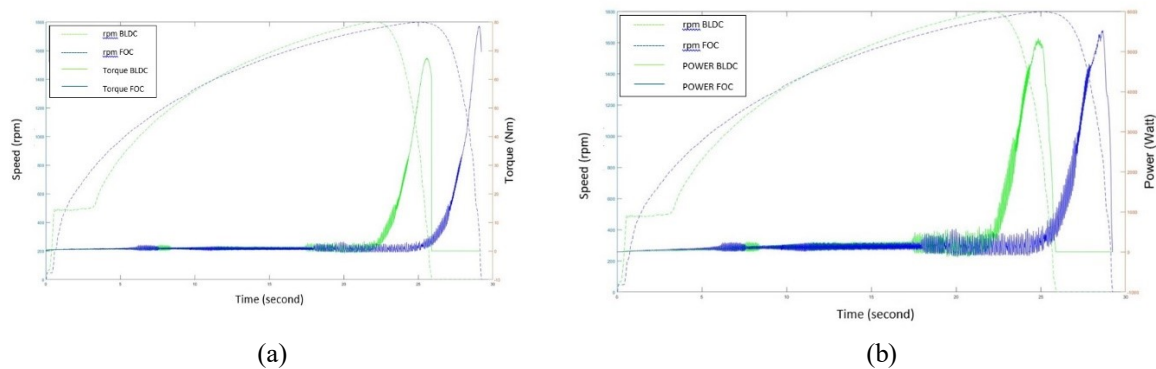


FIGURE 6. The result of the speed comparison between the trapezoidal controller and FOC controller: (a) Controller comparison based on torque (Nm), (b) Controller comparison based on power (Watt).

Next, experiments of speed step responses under initial load were carried out. Initially, a torque load was applied to the motor rotor through the dynamometer, then the motor rotor was rotated until it reached its steady-state speed. Fig. 7 shows the time acceleration results. The green lines are the results using the trapezoidal controller whereas the blue lines are the results using the FOC controller.

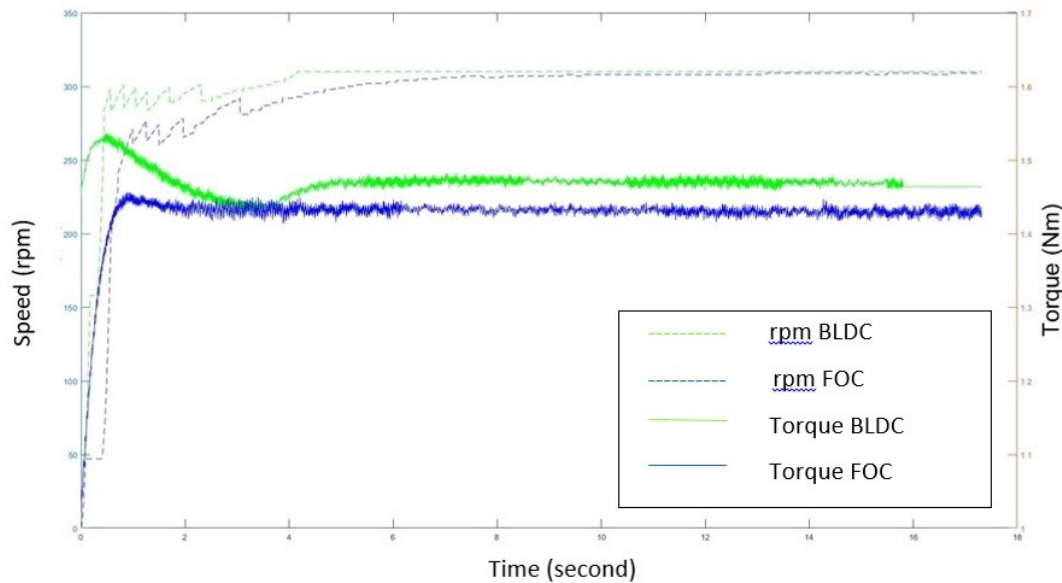


FIGURE 7. The result of torque comparison between the trapezoidal controller and FOC controller under initial load

Fig. 6 shows that the FOC controller has a smoother speed increase than the trapezoidal controller. When the speed increase during the test performance, the trapezoidal controller had a steady instantaneous condition at 500 rpm and raised again until the peak speed of 1800 rpm. Fig. 6 (a) shows that the trapezoidal controller generates the highest torque of 67 Nm at 300 rpm, while the FOC controller has its highest torque of 78 Nm at 324 rpm. Fig. 6 (b) shows that the trapezoidal controller generates the highest power of 5330 watt at 1074 rpm, while the FOC controller has its highest power of 5530 watt at 862 rpm. However, from Fig. 6 it is known that the time of the FOC controller to achieve the highest power and torque points take longer time than the trapezoidal controller.

Fig. 7 shows that the FOC controller has better torque response and more stable than the trapezoidal controller. But the trapezoidal controller has a faster rise time than the FOC controller. Based on the graph the trapezoidal controller is more responsive than the FOC controller.

Based on the results in Figs.6 and 7, when it is compared to the FOC controller, the trapezoidal controller is more responsive with faster speed response but produces higher torque overshoot and larger steady-state torque. In other words, the FOC controller provides smoother speed response with smaller torque overshoot and smaller steady-state torque than the trapezoidal controller.

CONCLUSION

We have compared the BLDC motor controller using trapezoidal control and field-oriented control through experiments. The experiment results indicate that field-oriented control produces smoother start-up speed with smaller torque delivery than the trapezoidal control. This result calls for deeper analysis to push performances of both types of the BLDC controllers to their limits and for further comparison in a possible higher stage of each controller's performance.

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